

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250279303

Kind Code

A1

Publication Date

September 04, 2025

Inventor(s)

MIKAMI; Kazuaki et al.

METHOD OF MANUFACTURING SEMICONDUCTOR DEVICE

Abstract

A method of manufacturing a semiconductor device includes a step of performing processing and a step of transporting. In the step of performing processing, processing for forming a semiconductor element is performed on a semiconductor substrate made from silicon carbide and having a diameter W of 200 mm or more. In the step of transporting, at least one of before and after processing, a semiconductor substrate is accommodated and transported in a FOUP, which is a transport carrier capable of accommodating a maximum of 25 semiconductor substrates.

Inventors: MIKAMI; Kazuaki (Tokyo, JP), NAKAMURA; Shotaro (Tokyo, JP), TSUCHIYA; Tsuyoshi (Tokyo, JP), YASUI; Satoshi (Tokyo, JP), KONISHI; Yasuo (Tokyo, JP)

Applicant: Mitsubishi Electric Corporation (Tokyo, JP)

Family ID: 96738167

Assignee: Mitsubishi Electric Corporation (Tokyo, JP)

Appl. No.: 18/991288

Filed: December 20, 2024

Foreign Application Priority Data

JP	2024-030897	Mar. 01, 2024
----	-------------	---------------

Publication Classification

Int. Cl.: H01L21/673 (20060101); H01L21/265 (20060101)

U.S. Cl.:

CPC H01L21/67383 (20130101); H01L21/26506 (20130101)

Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

[0001] The present disclosure relates to a method of manufacturing a semiconductor device.

Description of the Background Art

[0002] Switching to a semiconductor device made from silicon carbide (SiC), which has a lower loss than a semiconductor device made from silicon (Si) and can be expected to have higher reliability, is in progress. A semiconductor device made from Si is manufactured from a Si semiconductor substrate, and a semiconductor device made from SiC is manufactured from a SiC semiconductor substrate. A Si semiconductor substrate and a SiC semiconductor substrate are transported between processing apparatuses in a state of being accommodated in a transport carrier (also referred to as a transport container), and different processing is performed for each processing apparatus, so that a semiconductor element is formed on a semiconductor substrate (see, for example, Japanese Patent Application Laid-Open No. 2012-142620).

[0003] A SiC semiconductor substrate is different from a Si semiconductor substrate in not only a material but also a shape such as thickness in some cases, and a deformation amount of deflection such as warpage due to a weight of a SiC semiconductor substrate may be larger than a deformation amount of a Si semiconductor substrate. Further, a deformation amount becomes significant in a SiC semiconductor substrate having a diameter of 200 mm or more. For this reason, when a plurality of SiC semiconductor substrates having a diameter of 200 mm or more are transported in a state of being accommodated in a transport carrier, adjacent SiC semiconductor substrates may come into contact with each other during transportation, and a defect may occur.

[0004] On the other hand, in a conventional technique, a standard mechanical interface (SMIF) is used as a transport carrier that accommodates and transports a SiC semiconductor substrate. However, in an SMIF, since a housing for accommodating a SiC semiconductor substrate is small and a distance between accommodated SiC semiconductor substrates is short, there has been a problem that 25 semiconductor substrates cannot be transported as one unit due to the above-described defect, and manufacturability is poor.

SUMMARY

[0005] The present disclosure has been made in view of the above problem, and an object of the present disclosure is to provide a technique capable of appropriately transporting a semiconductor substrate.

[0006] A method of manufacturing a semiconductor device according to the present disclosure includes a step of preparing a semiconductor substrate made from silicon carbide and having a diameter W of 200 mm or more, a step of performing processing for forming a semiconductor element on the semiconductor substrate, and a step of accommodating and transporting the semiconductor substrate in a FOUP that is a transport carrier capable of accommodating a maximum of 25 of the semiconductor substrates at least one of before and after the processing.

[0007] The semiconductor substrate can be appropriately transported.

[0008] These and other objects, features, aspects and advantages of the present disclosure will become more apparent from the following detailed description of the present disclosure when taken in conjunction with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. 1 is a top view illustrating a configuration of a FOUP used in a method of

manufacturing a semiconductor device according to a first preferred embodiment;
[0010] FIG. **2** is a cross-sectional view illustrating a configuration of a part of a FOUP used in the method of manufacturing a semiconductor device according to the first preferred embodiment;
[0011] FIGS. **3** and **4** are top views each illustrating a configuration of a FOUP used in the method of manufacturing the semiconductor device according to the first preferred embodiment.
[0012] FIGS. **5** and **6** are flowcharts each showing the method of manufacturing a semiconductor device according to the first preferred embodiment;
[0013] FIG. **7** is a cross-sectional view illustrating the method of manufacturing of a semiconductor device according to the first preferred embodiment;
[0014] FIGS. **8** and **9** are two-dimensional maps each showing a deflection amount of a semiconductor substrate of a semiconductor device according to the first preferred embodiment;
[0015] FIG. **10** is a diagram illustrating a relationship between angles of an upper support point and a lower support point and a maximum deflection amount of a semiconductor substrate accommodated in an FOUP according to the first preferred embodiment;
[0016] FIG. **11** is a cross-sectional view illustrating a part of a configuration of a FOUP used in the method of manufacturing a semiconductor device according to a second preferred embodiment;
[0017] FIG. **12** is a cross-sectional view illustrating a part of a configuration of an FOUP used in the method of manufacturing a semiconductor device according to a third preferred embodiment;
and
[0018] FIG. **13** is a cross-sectional view illustrating a part of a configuration of a FOUP used in the method of manufacturing a semiconductor device according to a fourth preferred embodiment.

DESCRIPTION OF THE PREFERRED EMBODIMENTS

[0019] Hereinafter, a preferred embodiment will be described with reference to the attached drawings. Features described in the following preferred embodiments are examples, and all features are not necessarily essential. Further, in description below, similar constituent elements in a plurality of preferred embodiments are denoted by the same or similar reference numerals, and a different constituent element will be mainly described. Further, in description below, specific positions and directions such as “upper”, “lower”, “left”, and “right” do not need to coincide with actual positions and directions in practice.

First Preferred Embodiment

[0020] In a method of manufacturing a semiconductor device according to a first preferred embodiment, a front opening unified pod (FOUP) is used as a transport carrier that accommodates and transports a SiC semiconductor substrate. FIG. **1** is a top view illustrating a configuration of a FOUP **100** used in the method of manufacturing a semiconductor device according to the first preferred embodiment, and FIG. **2** is a cross-sectional view taken along line A-A' of FIG. **1**.

[0021] A semiconductor substrate **106** accommodated and transported in the FOUP **100** has a diameter W of 200 mm (8 inches) or more and is made from SiC (silicon carbide). Note that the semiconductor substrate **106** may include a wide band gap semiconductor other than SiC, such as gallium nitride (GaN), gallium oxide (Ga.sub.2O.sub.3), or diamond. In a semiconductor device formed of the semiconductor substrate **106** of a wide band gap semiconductor, stable operation under temperature and high voltage, and high switching speed can be achieved. Hereinafter, a case where the diameter W of the semiconductor substrate **106** is 200 mm will be mainly described. If the semiconductor substrate **106** is a 12-inch semiconductor substrate, the diameter W may be read as 300 mm. Note that the diameter W indicates a dimension when a semiconductor substrate manufacturer manufactures and ships a semiconductor substrate.

[0022] The semiconductor substrate **106** includes, for example, at least one of a normal semiconductor wafer and an epitaxial growth layer. Note that in the present description, for example, at least one of A, B, C, . . . , and Z means any one of all combinations obtained by extracting one or more types from groups of A, B, C, . . . , and Z. As illustrated in FIG. **2**, the semiconductor substrate **106** according to the first preferred embodiment may be a semiconductor

substrate **106a** having a small deflection amount or a semiconductor substrate **106b** having a large deflection amount.

[0023] The FOUP **100** is a transport carrier capable of accommodating a maximum of 25 of the semiconductor substrates **106**. As illustrated in FIGS. **1** and **2**, the FOUP **100** includes a partition portion **101**, an upper support point **102**, a lower support point **103**, a housing **104**, and a lid **105**. Note that in description below, in a case where the upper support point **102** and the lower support point **103** are not distinguished from each other, these may be collectively referred to as a “plurality of support points”.

[0024] The housing **104** has an opening portion that opens toward a first direction and through which the semiconductor substrate **106** is taken in and out. In a case where the semiconductor substrate **106** is not taken in and out, an opening portion of the housing **104** is closed by the lid **105**.

[0025] As illustrated in FIG. **2**, the partition portion **101** has an extending portion extending in a second direction orthogonal to the first direction, and both end portions in an extending direction of the extending portion are connected (fixed) to the inside of the housing **104** although not illustrated. The partition portion **101** has a protruding portion protruding in a third direction orthogonal to the first direction and the second direction. The partition portion **101** is configured to be capable of accommodating a plurality of the semiconductor substrates **106** in a state in which an outer peripheral portion of a plurality of the semiconductor substrates **106** up to 25 arranged at intervals of 10 mm in the second direction are partitioned by the protruding portion. Note that in the present description, a numerical value such as 10 mm may have a normal error.

[0026] As illustrated in FIG. **1**, in a case where an opening portion of the housing **104** is closed by the lid **105**, an end portion on the opening portion side of the semiconductor substrate **106** accommodated in the partition portion **101** abuts on the lid **105**. For this reason, in a case where an opening portion of the housing **104** is closed by the lid **105**, a position of the semiconductor substrate **106** in the first direction and the third direction is fixed by the partition portion **101** and the lid **105**. Note that, in FIG. **1**, the semiconductor substrate **106** is accommodated in the partition portion **101** such that a notch **108** of the semiconductor substrate **106** faces a direction opposite to the opening portion, but the semiconductor substrate **106** may be accommodated in the partition portion **101** such that the notch **108** faces a direction other than the above direction. Further, in a case where a plurality of the semiconductor substrates **106** are accommodated in the partition portion **101**, directions in which the notches **108** of a plurality of the semiconductor substrates **106** face do not need to be the same.

[0027] A plurality of support points include the upper support point **102** which is a support point closest to an opening portion and the lower support point **103** which is a support point farthest from the opening portion. As illustrated in FIG. **2**, a plurality of support points protrude in the second direction from the partition portion **101**. An interval between tips of support points adjacent to each other in the second direction is 10 mm, and a distance between support points adjacent to each other in the second direction is less than 10 mm. When the FOUP **100** is arranged in a posture in which the second direction is upward, a plurality of support points support both ends of each of a plurality of the semiconductor substrates **106** accommodated in the partition portion **101** along the second direction.

[0028] In a manufacturing process of a semiconductor device such as a power device, the semiconductor substrate **106** is transported from the FOUP **100** to a processing apparatus for processing by a wafer hand (not illustrated). Then, the processing apparatus performs processing for forming a semiconductor element to be a product on an upper surface that is a first main surface of the semiconductor substrate **106** or a lower surface that is a second main surface opposite to the first main surface. Then, the semiconductor substrate **106** is transported from the processing apparatus to the FOUP **100** by a wafer hand.

[0029] Note that, in order to suppress damage to a semiconductor element, a wafer hand that

transports the semiconductor substrate **106** desirably does not come into contact with a semiconductor element formation region, which is a region where the semiconductor element is formed. In order to realize this, an ineffective region where a semiconductor element is not formed for the wafer hand to come into contact with is preferably provided within 5.0 mm from an outer peripheral portion of the semiconductor substrate **106**. Further, in a case where processing of adjusting a shape of an outer peripheral portion of the semiconductor substrate **106** is performed as processing of the lower surface side of the semiconductor substrate **106**, a diameter W of a lower portion of the semiconductor substrate **106** may have size of 200 mm or less such as 198 mm ($=W-2$ mm), for example. For this reason as well, the ineffective region is preferably provided within 5.0 mm from an outer peripheral portion of the semiconductor substrate **106**. In a case where such an ineffective region is formed on the semiconductor substrate **106** having the diameter W of 200 mm, a range of a radius of 95 mm (diameter 190 mm) from the center of the semiconductor substrate **106** is an effective region where a semiconductor element is formed. That is, as illustrated in FIG. 1, a diameter W' of the effective region desirably satisfies $W-10\text{ mm} < W' < W$.

[0030] According to the FOUP **100** as described above, size of the housing **104**, and further, a distance between a plurality of support points in the second direction and the third direction can be made larger than that of a standard mechanical interface (SMIF). This makes it possible to make a distance between a plurality of the semiconductor substrates **106** accommodated in the partition portion **101** large, and thus, it is possible to transport a maximum of 25 of the semiconductor substrates **106** as one unit while suppressing contact between the semiconductor substrates **106**. As a result, manufacturability can be enhanced.

[0031] Further, since a plurality of support points provided in the partition portion **101** shift a deflected portion of the semiconductor substrate **106** in a direction opposite to a deflection direction, it is possible to prevent a wafer hand moving in the first direction from unintentionally coming into contact with the deflected portion of the semiconductor substrate **106**. Note that in FIG. 2, the semiconductor substrate **106a** having a small deflection amount and the semiconductor substrate **106b** having a large deflection amount are accommodated in the partition portion **101**, but the configuration may be such that one of the semiconductor substrate **106a** and the semiconductor substrate **106b** is accommodated in the partition portion **101**.

[0032] Further, the partition portion **101** of the FOUP **100** described above has a bent shape when viewed from the second direction as illustrated in FIG. 1, but is not limited to this, and may have, for example, a linear shape that is not bent. Further, in FIG. 1, the upper support point **102** and the lower support point **103** are provided on the same partition portion **101**, but the present invention is not limited to this. For example, as illustrated in FIG. 3, the partition portion **101** may include a first partition portion **101a** and a second partition portion **101b** located on the opposite side of an opening portion with respect to the first partition portion **101a**. Then, the upper support point **102** and the lower support point **103** may be provided on different ones of the partition portions **101** as in a configuration in which the upper support point **102** protrudes from the first partition portion **101a** and the lower support point **103** protrudes from the second partition portion **101b**.

[0033] Similarly to FIG. 1, FIG. 4 is a top view illustrating a configuration of the FOUP **100** according to the first preferred embodiment, and illustrates the arrangement of the upper support point **102** and the lower support point **103** viewed from the second direction. FIG. 4 illustrates a center O of the semiconductor substrate **106** accommodated in the partition portion **101**, an $X-X'$ line that is parallel to the third direction and passes through the center O , and a $Y-Y'$ line that is parallel to the first direction and passes through the center O .

[0034] In the first preferred embodiment, an angle **114** formed by a direction **111** from the center O of the semiconductor substrate **106** accommodated in the partition portion **101** toward the upper support point **102** closest to an opening portion among a plurality of support points and the third direction of the $X-X'$ line is 13° or more and 46° or less. Further, an angle **115** formed by a direction **112** from the center O of the semiconductor substrate **106** accommodated in the partition portion

101 toward the lower support point **103** farthest from an opening portion among a plurality of support points and the third direction of the X-X' line is 13° or more and 46° or less. Although details will be described later, according to such a configuration, it is possible to suppress deflection of the semiconductor substrate **106** accommodated in the FOUP **100**.

[0035] In the first preferred embodiment, a distance L (see FIG. 2) in the third direction between the upper support points **102** capable of supporting both ends of the semiconductor substrate **106** is larger than $W \times \cos 46^\circ$ mm and less than $W \times \cos 13^\circ$ mm. W' is a diameter of an effective region of the semiconductor substrate **106** (see FIG. 1). Note that although not illustrated, a distance in the third direction between the lower support points **103** capable of supporting both ends of the semiconductor substrate **106** is also larger than $W \times \cos 46^\circ$ mm and less than $W \times \cos 13^\circ$ mm.

Manufacturing Method

[0036] FIG. 5 is a flowchart showing the method of manufacturing a semiconductor device according to the first preferred embodiment. FIG. 5 illustrates a step of accommodating and transporting the semiconductor substrate **106** in the FOUP **100**, which is a transport carrier capable of accommodating a maximum of 25 of the semiconductor substrates **106**. The step of FIG. 5 is performed at least one of before and after processing described later for forming a semiconductor element on the semiconductor substrate **106**.

[0037] In Step S1, the FOUP **100** accommodating the semiconductor substrate **106** is transported from an automatic guided vehicle (AGV), an overhead hoist transport (OHT), or the like to a load port and loaded on the load port.

[0038] In Step S2, the load port reads a carrier ID attached to the FOUP **100**, and in Step S3, the load port moves and causes the FOUP **100** to be connected to a predetermined entrance and exit so that a wafer hand can transport the semiconductor substrate **106** accommodated in the FOUP **100** to a processing apparatus.

[0039] In Step S4, the lid **105** of the FOUP **100** is opened, and in Step S5, mapping for measuring a position and the number of the semiconductor substrates **106** accommodated in the FOUP **100** is performed. In Step S6, a wafer hand transports the semiconductor substrate **106** accommodated in the FOUP **100** from the FOUP **100** to the processing apparatus. In Step S7, the processing apparatus performs processing for forming a semiconductor element on the transported semiconductor substrate **106**. In Step S8, a wafer hand transports and accommodates the processed semiconductor substrate **106** from the processing apparatus to the FOUP **100**. Steps S6 to S8 are usually performed for each of the semiconductor substrates **106** accommodated in the FOUP **100**.

[0040] In Step S9, the lid **105** of the FOUP **100** is closed, and in Step S10, the load port releases connection between the FOUP **100** and the entrance and exit, and moves the FOUP **100** so that the AGV or the OHT can transport the FOUP **100**. In Step S11, the AGV or the OHT transports the FOUP **100**.

[0041] FIG. 6 is a flowchart illustrating the method of manufacturing a semiconductor device according to the first preferred embodiment, and illustrates an example of processing for forming a semiconductor element on the semiconductor substrate **106**. FIG. 7 is a cross-sectional view of the semiconductor substrate **106** illustrating the method of manufacturing a semiconductor device according to the first preferred embodiment. Hereinafter, a method of manufacturing an active region of a planar type metal oxide semiconductor field effect transistor (MOSFET) which is a type of semiconductor device will be described. However, the semiconductor device may be a trench type MOSFET, an insulated gate bipolar transistor (IGBT), a reverse conducting-IGBT (RC-IGBT), a Schottky barrier diode (SBD), a PN junction diode (PND), or the like. Note that in the first preferred embodiment, the active region is a semiconductor element formation region described above.

[0042] Processing for forming a semiconductor element on the semiconductor substrate **106** includes a semiconductor substrate preparing step (Step S21), an upper surface side p-type region forming step (Step S22), an upper surface side n-type region forming step (Step S23), an upper

surface side gate electrode forming step (Step S24), an upper surface side source electrode forming step (Step S25), an upper surface side protective film forming step (Step S26), a lower surface side grinding step (Step S27), a lower surface side drain electrode forming step (Step S28), an upper surface side protective film removing step (Step S29), and a dicing step (Step S30).

[0043] Note that processing performed before or after transport of the FOUP **100** does not need to include all the above steps. For example, in a case where it is necessary to transfer the semiconductor substrate **106** to a dedicated carrier in the dicing step in Step S30, the dicing step in Step S30 does not need to be included in processing performed before or after the FOUP **100** is transported. Hereinafter, a step in FIGS. **6** and **7** will be described.

[0044] In the semiconductor substrate preparing step (Step S21), a plurality of the semiconductor substrates **106** to be an n-type drift layer **200** in FIG. **7** are prepared. In this step, a semiconductor substrate made from silicon carbide and having the diameter W of 200 mm or more is prepared. In description of the manufacturing process below, for convenience, a configuration in which other semiconductor layers and electrodes are formed in the n-type drift layer **200** is also referred to as the semiconductor substrate **106**.

[0045] In the upper surface side p-type region forming step (Step S22), an ion implantation step and a heating step are performed. In the ion implantation step, a donor is ion-implanted into the upper surface side of the semiconductor substrate **106**. As the donor, for example, boron and aluminum are used. In the heating step, the semiconductor substrate **106** is heated to electrically activate the donor. By the above, a p-type region **201** is formed.

[0046] In the upper surface side n-type region forming step (Step S23), an exposure step, an etching step, an ion implantation step, and a heating step are performed. In the exposure step, photoresist having uniform thickness is applied to the upper surface side of the semiconductor substrate **106**. The photoresist may be either photosensitive or non-photosensitive. Then, a photoresist film is selectively exposed from the upper surface side of the semiconductor substrate **106** by using a photomask. In the etching step, the photoresist film is selectively removed by performing dry etching or wet etching on the photoresist film.

[0047] In the ion implantation step, an acceptor is ion-implanted into an upper surface of the p-type region **201** exposed from the photoresist film, and then a remaining photoresist film is removed. As the acceptor, for example, nitrogen, phosphorus, or the like is used. In the heating step, the semiconductor substrate **106** is heated to electrically activate an acceptor ion. By the above, an n-type region **202** is formed. As described above, by performing the exposure step and the etching step, a patterned impurity region and film can be formed, and an impurity region and film can be selectively formed on the semiconductor substrate **106**. Hereinafter, for convenience of description, description of removal of a remaining photoresist film, the exposure step, and the etching step may be omitted. A semiconductor element is formed on a semiconductor substrate by performing the ion implantation step and the heating step.

[0048] In the upper face side gate electrode forming step (Step S24), an oxide film forming step, a polysilicon film forming step, an exposure step, an etching step, and an oxide film forming step are performed. In the oxide film forming step, an oxide film **203** is formed on the p-type region **201** and the n-type region **202** by heating the semiconductor substrate **106** in atmosphere containing oxygen. In the polysilicon film forming step, a polysilicon film doped with n-type or p-type impurities is deposited by chemical vapor deposition (CVD) or the like. In the exposure step, photoresist having uniform thickness is applied to the upper surface side of the semiconductor substrate **106**. The photoresist may be either photosensitive or non-photosensitive. Then, a photoresist film is selectively exposed from the upper surface side of the semiconductor substrate **106** by using a photomask.

[0049] In the etching step, the photoresist film is selectively removed by performing dry etching or wet etching on the photoresist film. Then, a polysilicon film exposed from the photoresist film is removed to form a gate electrode **204** facing an upper portion of the p-type region **201** with the

oxide film **203** interposed between them. In the oxide film forming step, an oxide film **205** covering the gate electrode **204** is formed by heating the semiconductor substrate **106** in atmosphere containing oxygen or performing CVD or the like. After the above, the oxide films **203** and **205** between the gate electrodes **204** are removed to expose the n-type region **202**.

[0050] In the upper surface side source electrode forming step (Step S25), a source electrode **206** connected to the n-type region **202** is formed on the upper surface side of the semiconductor substrate **106** by a sputtering apparatus or the like. A material of the source electrode **206** is, for example, nickel. In order to reduce contact resistance of the source electrode **206**, the source electrode **206** may be formed after heat treatment and a silicide process are performed on an upper surface of the n-type region **202**.

[0051] In the upper surface side protective film forming step (Step S26), a protective film **207** is formed on the source electrode **206**, that is, on the upper surface side of the semiconductor substrate **106**. As the protective film **207**, for example, tape, a glass plate, or the like is used, but one having small thickness is more desirable from the viewpoint of cost. In the steps so far, since thickness of the semiconductor substrate **106** is sufficiently large, deflection of the semiconductor substrate **106** in the FOUP **100** is about a fraction of a millimeter.

[0052] In the lower surface side grinding step (Step S27), front and back surfaces of the semiconductor substrate **106** are reversed, the protective film **207** on the upper surface side of the semiconductor substrate **106** is adsorbed on a stage of a grinding apparatus, and the lower surface side of the semiconductor substrate **106** is ground so that the n-type drift layer **200** has thickness of several 100 μm . Note that, for example, Japanese Patent No. 6194210 proposes a grinding method for leaving a rim in order to suppress deflection of the semiconductor substrate **106**. However, in a case where the semiconductor substrate **106** is made from a hard brittle material such as SiC, it is desirable to grind the semiconductor substrate **106** so that thickness becomes uniform as a whole in order to suppress abrasion of a grindstone, occurrence of cracking of the semiconductor substrate **106**, and the like. Since a damaged layer remains on a surface layer on the lower surface side of the semiconductor substrate **106** after grinding, the damaged layer may be removed by etching.

[0053] The semiconductor substrate **106** after grinding is easily deformed, and as in the semiconductor substrate **106a** accommodated in the FOUP **100** of FIG. 2, deflection in the second direction due to its own weight becomes maximum at the center.

[0054] In the lower surface side drain electrode forming step (Step S28), as illustrated in FIG. 7, a drain electrode **208** is formed on the lower surface side of the semiconductor substrate **106** by using a sputtering apparatus or the like. A material of the drain electrode **208** is, for example, nickel, titanium, or the like. In order to reduce contact resistance of the drain electrode **208**, the drain electrode **208** may be formed after heat treatment and a silicide process are performed on a lower surface of the n-type drift layer **200** (corresponding to an upper surface in Step S28 in FIG. 7). For the heat treatment, for example, a laser is used.

[0055] In the upper surface side protective film removing step (Step S29), front and back surfaces of the semiconductor substrate **106** are reversed, and the protective film **207** provided on the upper surface side of the semiconductor substrate **106** is removed. In a case where the protective film **207** is tape, the tape is physically peeled off the semiconductor substrate **106**, and tape adhesive residue is removed by wet etching or the like. Note that in a case where heat-resistant temperature of the protective film **207** is lower than rising temperature in the lower surface side drain electrode forming step (Step S28), the lower surface side drain electrode forming step (Step S28) and the upper surface side protective film removing step (Step S29) are preferably performed in the reversed order.

[0056] In the dicing step (Step S30), each semiconductor element is cut out from the semiconductor substrate **106**. Through the above steps, a semiconductor element is manufactured as a semiconductor device. However, a semiconductor device as a target of the method of manufacturing a semiconductor device may be a semiconductor module in which a semiconductor

element is provided with, for example, a wire or the like and then sealed with sealing resin, or may be a device in which the semiconductor module is further incorporated.

Summary of First Preferred Embodiment

[0057] According to the method of manufacturing a semiconductor device according to the first preferred embodiment as described above, the semiconductor substrate **106** is accommodated and transported in the FOUP **100** that can accommodate a maximum of 25 of the semiconductor substrates **106** having the diameter W of 200 mm or more made from silicon carbide. According to such a configuration, it is possible to transport a maximum of 25 of the semiconductor substrates **106** as one unit while suppressing contact between the semiconductor substrates **106**, and thus, it is possible to enhance manufacturability.

[0058] The above effect is particularly effective in a case where a maximum of 25 of the semiconductor substrates **106** are transported after the lower surface side grinding step (Step S27) in which a deflection amount of the semiconductor substrates **106** is large. Further, in the upper surface side protective film forming step (Step S26), the above effect is particularly effective in a case where tape that increases a deflection amount of the semiconductor substrate **106** is used as the protective film **207**. Further, in a case where a glass plate is used as the protective film **207** in the upper surface side protective film forming step (Step S26), a deflection amount of the semiconductor substrate **106** can be reduced.

[0059] Further, in the first preferred embodiment, since a plurality of support points are provided in the partition portion **101**, it is possible to prevent a wafer hand from unintentionally coming into contact with a deflected portion of the semiconductor substrate **106**.

[0060] Further, in the first preferred embodiment, the angle **114** formed by the direction **111** from the center O of the semiconductor substrate **106** accommodated in the partition portion **101** toward the upper support point **102** closest to an opening portion among a plurality of support points and the third direction of the X-X' line is 13° or more and 46° or less. Further, an angle **115** formed by a direction **112** from the center O of the semiconductor substrate **106** accommodated in the partition portion **101** toward the lower support point **103** farthest from an opening portion among a plurality of support points and the third direction of the X-X' line is 13° or more and 46° or less. Hereinafter, an effect of this configuration will be described.

[0061] FIGS. **8** and **9** are two-dimensional maps illustrating a deflection amount of the semiconductor substrate **106** in a case where the semiconductor substrate **106** in the lower surface side drain electrode forming step (Step S28) is supported in the FOUP **100** at a plurality of support points having the angles **114** and **115** of 30°. FIG. **8** is a two-dimensional map when the semiconductor substrate **106** is accommodated in the partition portion **101** such that the upper surface side faces TOP (the upper side in FIG. 2). FIG. **9** is a two-dimensional map when the semiconductor substrate **106** is accommodated in the partition portion **101** such that the upper surface side faces BOTTOM (the lower side in FIG. 2). FIG. **8** shows that a deflection amount has a maximum value (3.2 mm) at a center portion parallel to the first direction in the semiconductor substrate **106**. On the other hand, FIG. **9** shows that a deflection amount has a maximum value (2.0 mm) at both end portions parallel to the third direction in the semiconductor substrate **106**.

[0062] FIG. **10** is a diagram illustrating a relationship between the angles **114** and **115** of the upper support point **102** and the lower support point **103** in FIG. 4 and a maximum deflection amount of the semiconductor substrate **106** accommodated in the FOUP **100**. A plot **301** shows a maximum deflection amount (hereinafter, also referred to as a “TOP direction maximum deflection amount”) when the semiconductor substrate **106** is accommodated such that an upper surface faces TOP as in a deflection amount of FIG. **8**. A plot **302** shows a maximum deflection amount (hereinafter, also referred to as a “BOTTOM direction maximum deflection amount”) when the semiconductor substrate **106** is accommodated such that an upper surface faces BOTTOM as in a deflection amount of FIG. **9**. As can be seen from the plots **301** and **302**, the TOP direction maximum deflection amount is larger than the BOTTOM direction maximum deflection amount.

[0063] Here, in the description of steps S27 to S29, front and back surfaces of the semiconductor substrate **106** are reversed, but when such reversal is performed, throughput is reduced. In view of the above, hereinafter, a case where the semiconductor substrate **106** is processed and transported without reversal of the semiconductor substrate **106** in Steps S27 to S29 will be assumed. In this case, since the semiconductor substrate **106** is accommodated in the partition portion **101** such that the upper surface side faces TOP (the upper side in FIG. 2), the TOP direction maximum deflection amount is assumed.

[0064] In order for a wafer hand having thickness of 5.0 mm in the second direction to move to a mounting position of the semiconductor substrate **106** without coming into contact with the semiconductor substrate **106** and the partition portion **101**, a gap of 1.0 mm is required between the wafer hand and the semiconductor substrate **106**. Considering that the semiconductor substrates **106a** and **106b** are accommodated in the FOUP **100** as illustrated in FIG. 2, since an interval between tips of support points adjacent in the second direction is 10 mm, a deflection amount of the semiconductor substrate **106** can be allowed up to 4.0 mm.

[0065] In FIG. 10, from an intersection of the plot **301** and the TOP direction maximum deflection amount of 4.0 mm, the angles **114** and **115** can be allowed up to 13°. On the other hand, when the angle **114** (angle **115**) in FIG. 4 increases, a distance in the third direction between the upper support points **102** (between the lower support points **103**) decreases. Assuming that width of a wafer hand in the third direction is 170 mm or less, the angles **114** and **115** can be allowed up to 46°. In the first preferred embodiment, since the angles **114** and **115** of the upper support point **102** and the lower support point **103** is 13° or more and 46° or less, a maximum deflection amount of the semiconductor substrate **106** can be appropriately suppressed.

Second Preferred Embodiment

[0066] FIG. 11 is a cross-sectional view illustrating the upper support point **102** and the lower support point **103** according to the second preferred embodiment as viewed from the third direction. In the second preferred embodiment, as illustrated in FIG. 11, the upper support point **102** is higher than the lower support point **103** by, for example, 0.1 mm in the second direction. According to such a configuration, it is possible to suppress deflection of the semiconductor substrate **106** in which the TOP direction maximum deflection amount exists on the opening portion side of the FOUP **100** as illustrated in FIG. 8.

Third Preferred Embodiment

[0067] FIG. 12 is a cross-sectional view illustrating a plurality of support points according to a third preferred embodiment as viewed from the first direction. Although not illustrated, a configuration obtained by horizontally reversing the configuration of FIG. 12 is provided on the right side of FIG. 12. In the third preferred embodiment, when viewed from the first direction, a tip of a first set of support points **401** among a plurality of support points is higher than a tip of a second set of support points **402** further on the inner side than the first set of the support points **401** among a plurality of support points. According to such a configuration, the semiconductor substrates **106** having the different diameters W can be accommodated in the same FOUP **100**, and deflection of the semiconductor substrates **106** having the different diameters W can be suppressed.

Fourth Preferred Embodiment

[0068] FIG. 13 is a cross-sectional view illustrating the upper support point **102** according to the fourth preferred embodiment as viewed from the first direction. In the fourth preferred embodiment, as illustrated in FIG. 12, the upper support point **102** is in surface contact with the semiconductor substrate **106** accommodated in the partition portion **101**. Further, although not illustrated, the lower support point **103** is also in surface contact with the semiconductor substrate **106** accommodated in the partition portion **101**. According to such a configuration, as compared with a configuration in which the upper support point **102** and the lower support point **103** are in point contact with the semiconductor substrate **106**, stress on the semiconductor substrate **106** can be reduced, and deflection of the semiconductor substrate **106** can be suppressed.

Variation

[0069] In FIGS. 1, 2, 4, 12, and 13, the partition portion **101** may be detachable. According to such a configuration, when the partition portion **101** is replaced with a partition portion for supporting a semiconductor substrate having the diameter W of 300 mm, a semiconductor substrate having the diameter W of 300 mm can be accommodated in the same housing **104**. Therefore, the number of the FOUPs **100** to be prepared can be reduced, and capacity of a stocker for storing the FOUPs **100** can be reduced.

[0070] Note that the preferred embodiments and the variations can be freely combined, and the preferred embodiments and the variations can be appropriately modified or omitted.

[0071] Hereinafter, various aspects of the present disclosure will be collectively described as an appendix.

Appendix 1

[0072] A method of manufacturing a semiconductor device, the method including: [0073] a step of preparing a semiconductor substrate made from silicon carbide and having a diameter W of 200 mm or more; [0074] a step of performing processing for forming a semiconductor element on the semiconductor substrate; and [0075] a step of accommodating and transporting the semiconductor substrate in a FOUP that is a transport carrier capable of accommodating a maximum of 25 of the semiconductor substrates at least one of before and after the processing.

Appendix 2

[0076] The method of manufacturing a semiconductor device according to Appendix 1, in which [0077] the processing includes at least one of: [0078] a step of forming a plurality of impurity regions in the semiconductor substrate; [0079] a step of forming a protective film on a first main surface of the semiconductor substrate; and [0080] a step of grinding a second main surface opposite to the first main surface of the semiconductor substrate after the protective film is formed.

Appendix 3

[0081] The method of manufacturing a semiconductor device according to Appendix 2, in which [0082] the protective film is a glass plate.

Appendix 4

[0083] The method of manufacturing a semiconductor device according to Appendix 2, in which [0084] the protective film is tape.

Appendix 5

[0085] The method of manufacturing a semiconductor device according to any one of Appendices 1 to 4, in which [0086] the FOUP includes: [0087] a housing having an opening portion that opens toward a first direction and through which the semiconductor substrate is taken in and out; [0088] a partition portion connected to an inside of the housing and capable of partitioning and accommodating an outer peripheral portion of a plurality of the semiconductor substrates up to 25 arranged at an interval of 10 mm in a second direction orthogonal to the first direction; and [0089] a plurality of support points protruding from the partition portion in the second direction and capable of supporting each of a plurality of the semiconductor substrates accommodated in the partition portion along the second direction, [0090] when viewed from the second direction, an angle formed by a direction from a center of the semiconductor substrate accommodated in the partition portion toward the support point closest to the opening portion among a plurality of the support points and a third direction orthogonal to the first direction and the second direction is 13° or more and 46° or less, [0091] $W-10\text{ mm} < W' < W$ holds for a diameter W' of an effective region of the semiconductor substrate, and [0092] a distance in the third direction between support points capable of supporting the semiconductor substrate among a plurality of the support points is larger than $W \times \cos 46^\circ$ mm and less than $W \times \cos 13^\circ$ mm.

Appendix 6

[0093] The method of manufacturing a semiconductor device according to Appendix 5, in which [0094] the partition portion includes a first partition portion and a second partition portion located

on a side opposite to the opening portion with respect to the first partition portion, and [0095] the support point closest to the opening portion among a plurality of the support points protrudes from the first partition portion, and [0096] the support point farthest from the opening portion among a plurality of the support points protrudes from the second partition portion.

Appendix 7

[0097] The method of manufacturing a semiconductor device according to Appendix 5 or 6, in which [0098] an angle formed by the third direction and a direction from a center of the semiconductor substrate accommodated in the partition portion toward the support point farthest from the opening portion among a plurality of the support points when viewed from the second direction is 13° or more and 46° or less.

Appendix 8

[0099] The method of manufacturing a semiconductor device according to any one of Appendices 5 to 7, in which [0100] a support point closest to the opening portion among a plurality of the support points is higher in the second direction than the support point farthest from the opening portion among the plurality of support points.

Appendix 9

[0101] The method of manufacturing a semiconductor device according to any one of Appendices 5 to 8, in which [0102] a tip of a first set of the support points among a plurality of the support points is higher than a tip of a second set of the support points further on an inner side than the first set of the support points among a plurality of the support points when viewed from the first direction.

Appendix 10

[0103] The method of manufacturing a semiconductor device according to any one of Appendices 5 to 9, in which [0104] a plurality of the support points are in surface contact with the semiconductor substrate accommodated in the partition portion.

Appendix 11

[0105] The method of manufacturing a semiconductor device according to any one of Appendices 5 to 10, in which [0106] the partition portion is replaceable.

[0107] While the disclosure has been shown and described in detail, the foregoing description is in all aspects illustrative and not restrictive. It is therefore understood that numerous modifications and variations can be devised.

Claims

1. A method of manufacturing a semiconductor device, the method comprising: a step of preparing a semiconductor substrate made from silicon carbide and having a diameter W of 200 mm or more; a step of performing processing for forming a semiconductor element on the semiconductor substrate; and a step of accommodating and transporting the semiconductor substrate in a FOUP that is a transport carrier capable of accommodating a maximum of 25 of the semiconductor substrates at least one of before and after the processing.
2. The method of manufacturing a semiconductor device according to claim 1, wherein the processing includes at least one of: a step of forming a plurality of impurity regions in the semiconductor substrate; a step of forming a protective film on a first main surface of the semiconductor substrate; and a step of grinding a second main surface opposite to the first main surface of the semiconductor substrate after the protective film is formed.
3. The method of manufacturing a semiconductor device according to claim 2, wherein the protective film is a glass plate.
4. The method of manufacturing a semiconductor device according to claim 2, wherein the protective film is tape.
5. The method of manufacturing a semiconductor device according to claim 1, wherein the FOUP includes: a housing having an opening portion that opens toward a first direction and through

which the semiconductor substrate is taken in and out; a partition portion connected to an inside of the housing and capable of partitioning and accommodating an outer peripheral portion of a plurality of the semiconductor substrates up to 25 arranged at an interval of 10 mm in a second direction orthogonal to the first direction; and a plurality of support points protruding from the partition portion in the second direction and capable of supporting each of the plurality of semiconductor substrates accommodated in the partition portion along the second direction, when viewed from the second direction, an angle formed by a direction from a center of the semiconductor substrate accommodated in the partition portion toward the support point closest to the opening portion among the plurality of support points and a third direction orthogonal to the first direction and the second direction is 13° or more and 46° or less, $W-10\text{ mm} < W' < W$ holds for a diameter W' of an effective region of the semiconductor substrate, and a distance in the third direction between support points capable of supporting the semiconductor substrate among the plurality of support points is larger than $W \times \cos 46^\circ$ mm and less than $W \times \cos 13^\circ$ mm.

6. The method of manufacturing a semiconductor device according to claim 5, wherein the partition portion includes a first partition portion and a second partition portion located on a side opposite to the opening portion with respect to the first partition portion, and the support point closest to the opening portion among the plurality of support points protrudes from the first partition portion, and the support point farthest from the opening portion among the plurality of support points protrudes from the second partition portion.

7. The method of manufacturing a semiconductor device according to claim 5, wherein an angle formed by the third direction and a direction from a center of the semiconductor substrate accommodated in the partition portion toward the support point farthest from the opening portion among the plurality of support points when viewed from the second direction is 13° or more and 46° or less.

8. The method of manufacturing a semiconductor device according to claim 5, wherein a support point closest to the opening portion among the plurality of support points is higher in the second direction than the support point farthest from the opening portion among the plurality of support points.

9. The method of manufacturing a semiconductor device according to claim 5, wherein a tip of a first set of the support points among the plurality of support points is higher than a tip of a second set of the support points further on an inner side than the first set of the support points among the plurality of support points when viewed from the first direction.

10. The method of manufacturing a semiconductor device according to claim 5, wherein the plurality of support points are in surface contact with the semiconductor substrate accommodated in the partition portion.

11. The method of manufacturing a semiconductor device according to claim 5, wherein the partition portion is replaceable.
